Naming Compounds
&
Writing Formula

Metallic substances
The name of the metallic substance is the same as the element.
The formula is the element symbol.
(sodium – Na, iron-Fe, copper-Cu etc)

Ionic compounds
Rules for naming:
1. The metal or positive ion is named first and keeps its normal name.
2. The non-metal or negative ion is named second and has the end of its name written as –ide, -ate or –ite.
3. If the metal ion has several valencies, indicate its valency using roman numerals.

(sodium chloride, ammonium sulfate, iron sulfite etc)

Writing formula:
1. The positive ion is written first.
2. The negative ion is written second.
3. The number of each ion is adjusted so that the total positive charge equals the total negative charge. Subscripts are used to indicate the number of each ion present.

(sodium chloride-NaCl, ammonium phosphate- $(NH_4)_3PO_4$, iron(II) hydroxide- $Fe(OH)_2$)

Covalent Compounds
Rules for naming
1. The name of the element closer to the bottom left side of the periodic table is named first and keeps its normal name.
2. The name of the second element has its name changed to end in –ide e.g. sulphide, chloride, oxide, nitride etc
3. Prefixes are used to indicate the number of atoms of each element.
4. If there is only one atom of the first element then no prefix is used. For example, CO_2 – carbon dioxide
5. Molecules composed of one element are given the name of that element. For example, H_2-hydrogen, O_2-oxygen, Cl_2- chlorine

Number of atoms	Prefix
1	mono-
2	di-
3	tri-
4	tetra-
5	penta-

Number of atoms	Prefix
6	hexa-
7	hepta-
8	octa-
9	nona-
10	deca-

Examples:

1. Zn - zinc

2. sodium iodide - NaI
   calcium nitrate – $Ca(NO_3)_2$

3. $CuSO_4$ – copper(II) sulfate
   Fe_2O_3 – iron(iii) oxide

4. nitrogen trifluoride – NF_3
   Tetraphosphorus decaoxide – P_4O_{10}
   Nitrogen gas – N_2

5. SO_3 – sulphur trioxide
   S_2Cl_4 –disulfur tetrachloride

Exercises:

1. Name the following ionic compounds.

   a) ZnS

   b) $Mg(OH)_2$

   c) $CuCO_3$

   d) $AlCl_3$

   e) $BaSO_4$

2. Write the correct formula for the following compounds.

   a) copper(II) chloride

   b) calcium hydroxide

   c) ammonium carbonate

   d) tin(II) sulfate

   e) chromium(III) phosphate

   f) silver chloride

3. Name the following covalent compounds.

   a) Br_2

   b) P_2O_5

   c) SO_2

   d) HCl

   e) N_2O_4

4. Write the formula for the covalent compounds written below.

   a) oxygen dichloride

   b) carbon tetrafluoride

   c) hydrogen iodide

   d) silicon dioxide

   e) carbon monoxide